Saurav Dosi

sauravdosi@gmail.com sdo4011@med.cornell.edu saurav.dosi@utdallas.edu
sauravdosi.com linkedin.com/in/sauravdosi github.com/sauravdosi

Research Interests

Computer Vision • Diffusion Models • Robotics • Multimodal AI • Large Language Models

Education

The University of Texas at Dallas Dallas, TX
Master of Science, Computer Science Aug 2023 – *Expected:* May 2026
(Intelligent Systems Track)

- *Research Advisor*: Dr. Yu Xiang; (IntelX) Intelligent Robot Exploration, IRVL
- **GPA:** 3.98 / 4.0; *Expected Summa Cum Laude*
- **Award:** Jonsson School Dean's Graduate Scholarship, 2023

Indian Institute of Technology, Dharwad Dharwad, India
Bachelor of Technology, Mechanical Engineering Jul 2017 – Jun 2021
(Computer Science Minor)

- *Research Advisor*: Dr. Samarth Raut; Defended Thesis, Published in Springer 2022
- **GPA:** 8.55 / 10.0
- **Award:** Department Rank 2

Publications

Webpage

Springer 2022 Segregation of Areca Nuts Using Three Band Photometry and Deep Neural Network
Saurav Dosi*, Vamsi Bala*, Samarth S. Raut, and D. Narasimha
Conference on Soft Computing and its Engineering Applications, May 2022 [Paper]

In Prep IntelX: Intelligent Exploration using 2D Occupancy Coverage Prediction in Indoor Environments
Saurav Dosi, Itay Kadosh, Sai Haneesh Allu, and Yu Xiang

Medical Physics Submitted Inter-crystal Scatter Recovery of Prism-PET Prototype Scanner using Diffusion Model
Wanbin Tan, **Saurav Dosi**, Yixin Li, Zipai Wang, Eric Petersen, Xinjie Zeng, Soroush Shabani Sichani, and Amir H. Goldan

SNMMI 2026 Submitted D3Recovery: Diffusion-guided Tri-model Recovery for Prism-PET III Multiplexed Brain Scanner
Saurav Dosi, Wanbin Tan, Yixin Li, Zipai Wang, Eric Petersen, Xinjie Zeng, Soroush Shabani Sichani, and Amir H. Goldan

SNMMI 2026 Submitted MRI-less MK-tauPET staging using 3D Diffusion for Alzheimer's Disease
Saurav Dosi, Wanbin Tan, Yixin Li, Zipai Wang, Eric Petersen, Xinjie Zeng, Soroush Shabani Sichani, and Amir H. Goldan

(* denotes equal contribution.)

Research Experience

Weill Cornell Medicine, Cornell University New York, NY
Research Assistant, X-ray and Emission Imaging Lab (XEIL) Jul 2025 – Present

Advised by: Dr. Amir H. Goldan

- Proposed **D3Recovery** involving Scatter Diffusion, CrystalAttentionNet and GuidedDepthNet for the recovery of PET scans to remove effects of inter-crystal scatter noise.
- Devising a 3D Diffusion model for Low-dose MRI-less TauPET screening for Alzheimer's disease.
- Developing a multimodal Diffusion model to synthesize PET scans from neural temporal data (EEG).
- Conducted literature survey on wearable foundation models e.g. Apple's Wearable Behavior Model.
- **Keywords**: *Computer Vision, Diffusion Models, Multimodal AI, Medical Imaging, Transformers*

The University of Texas at Dallas — Dallas, TX
Research Assistant Volunteer, Intelligent Robotics and Vision Lab (IRVL) — Oct 2023 – Present
Advised by: Dr. Yu Xiang

- Proposed **IntelX**, a Robot Exploration Policy:
  - To intelligently explore unknown indoor environments solely using 2D SLAM scans.
  - Designed a novel UNet Segmentation Regression approach for map coverage prediction and an RL policy for global planning.
  - Trained on 1M samples from 18 long horizon maps, outperformed Frontier Exploration (AutoXSemMap) by 20-25% in path length and time.
- Developed SLAM-agnostic PointNav using Soft Actor-Critic for efficient goal-reaching.
- Engineered Python interface in NVIDIA Isaac Sim for 3+ lab projects, enabling robot navigation environment setup and controller/sensor simulation on Fetch robot.
- **Keywords:** *Robot Exploration, SLAM, Image Segmentation, Reinforcement Learning, Simulation*

Industry Experience

Boehringer Ingelheim — Ridgefield, CT
Data Scientist Co-Op — Aug 2025 – Present

- Developed a vision-language Chatbot harnessing RAG and Agentic Web Search (GPT-5) for the company's Sustainability policy.
- Utilized GPT-4o to enable Excel processing along with Visual Analytics using CLIP image retrieval.
- Researched hierarchical planner-solver agent framework with multi-hop reasoning.
- **Keywords:** *Vision-Language Models, RAG, Agents, Reasoning, CLIP, Large Language Models*

ISN Software Corporation — Dallas, TX
Data Science (NLP) Intern — Jun 2024 – Aug 2024

- Proposed Transformer-backed **IntelliMatch**, Intelligent Name Matching:
  - A multifaceted solution for company matching, grouping, and industry prediction.
  - Utilized Self-Attention layers, and k-fold TF-IDF, Trie and NumPy vectorization.
- **Keywords:** *Transformers, Self-Attention, NLP, Entity Matching, Text Classification*

Quantrium AI — Chennai, India
Machine Learning Engineer — Jul 2021 – Jul 2023

- Led a team of 3 for Intelligent Shelf Auditor using RetinaNet and FRCNN, achieving 95% accuracy.
- Led development of Sportify using Multi-Object Tracking in Soccer analytics based on ByteTrack.
- Spearheaded and developed Document Intelligence with 98.3% accuracy using SpaCy BERT NER.
- **Keywords:** *Object Detection, Multi-Object Tracking, Video Analytics, NER, Document AI, OCR*

Mirrag AI — Mumbai, India
Artificial Intelligence Engineer — Apr 2021 – Jun 2021

- Developed 90% accurate real-time License Plate Recognizer using YOLOv5 and PaddleOCR.
- Engineered a vision-based Early-stage Fire Detection method outperforming traditional detectors.
- **Keywords:** *Object Detection, Real-time Vision, YOLO, OCR, Video Processing*

Express Analytics — Pune, India
Data Science Intern — Sep 2020 – Jan 2021

- Built 93% accurate Sentiment Tool with emoji support using Flair and Flask.
- Researched Fake review detection algorithms.
- **Keywords:** *Sentiment Analysis, NLP, Text Classification, Anomaly Detection, SpaCy*

Teaching Experience

The University of Texas at Dallas Dallas, TX
Teaching Assistant Sep 2024 – May 2025
Supervisor: Dr. Sridhar Alagar
CS 3345: Data Structures and Algorithms (140+ students in Fall 2024, 80+ in Spring 2025)

- Delivered an in-person lecture on competitive challenges on linked lists to 80 undergraduate juniors.
- Automated code grading utilizing LLMs, resulting in a 50% time savings.
- Conducted 1-on-1 mentoring during office hours and led review sessions.
- **Keywords:** *LLM-assisted Grading, Python, Automation, Algorithms, Data Structures*

Reading Group Presentations

XEIL Mediffuse: CT2MRI Diffusion-Driven Cross-Modal Translation
Saurav Dosi, Animesh Maheshwari, Pratiksha Aigal, Varad Abhyankar, Yapeng Tian

IRVL OmniManip - Towards General Robotic Manipulation via Object-Centric Interaction Primitives
Mingjie Pan, Jiyao Zhang, Tianshu Wu, Yinghao Zhao, Wenlong Gao, Hao Dong
Computer Vision and Pattern Recognition (CVPR) 2025 Highlight

IRVL NoMaD - Goal Masked Diffusion Policies for Navigation and Exploration
Ajay Sridhar, Dhruv Shah, Catherine Glossop, Sergey Levine
IEEE International Conference on Robotics and Automation (ICRA) 2024 Best Paper Award Winner

IRVL Real-world Mobile Manipulation with RL - Continuously Improving Mobile Manipulation with Autonomous Real-World RL
Russell Mendonca1, Emmanuel Panov, Bernadette Bucher, Jiuguang Wang, Deepak Pathak

Projects

Mediffuse: Diffusion-driven CT to MRI Translation UTD
Computer Vision Research Project, CS 6384 [Report] March 2025 – May 2025
Instructor: Dr. Yapeng Tian

- Fine-tuned a CT2MRI Diffusion pipeline with 0.75 SSIM InstructPix2Pix, ControlNet++, SD1.5.
- Achieved 6s inference time, out-of-distribution SSIM ≥ 0.4 and tumor detection SSIM ≥ 0.5.
- **Keywords:** *Diffusion Models, Img2Img Translation, Medical Imaging, InstructPix2Pix, ControlNet*

DOG: Dynamic Object Grasping UTD
Robotics Research Project, CS 6301 [Report] Sep 2024 – Dec 2024
Instructor: Dr. Yu Xiang

- Developed robotic grasping system for moving objects using real-time Recursive Least Squares motion prediction on deterministic 2D paths.
- **Keywords:** *Robotic Manipulation, Motion Prediction, Grasping, Control Systems*

CometSeek: AI Chatbot for UTD Assistance UTD
Reasoning LLM and Fine-tuning Project March 2025 – April 2025

- Built an 85% accurate Reasoning chatbot for UTD assistance using a fine-tuned DeepSeek R1 distilled LLaMA 8B model with ∼15s response time.
- Surpassed LoRA UTD mascot-like tone fine-tuning limitations via prompt engineering and GRPO.
- Enhanced accuracy by implementing Agentic RAG reranking, along with query rephrasing, query tools, and hybrid information retrieval.
- **Keywords:** *LLM Fine-tuning, RAG, Agentic AI, Reasoning Models, RLHF, Word Embeddings*

Virtual Try-on using Diffusion Models UTD
Research Project Oct 2024 – Nov 2024

- Ensembled IDM-VTON and OOTDiffusion to achieve 85%+ CLIP and SSIM scores on OOD images.
- **Keywords:** *Diffusion Models, Model Ensembling, CLIP Embeddings, Multimodal AI*

IntelliMatch: Intelligent Name Matching ISN
Internship Project Jun 2024 – Aug 2024

- Built a pipeline integrating Transformers to prioritize keywords by semantic role for entity resolution.
- **Keywords:** *Transformers, Self-Attention, Entity Recognition, Semantic Matching, TensorFlow*

InvestAID: AI powered Investment Dashboard UTD
HackUTD 2023 Project Nov 2023

- Developed a ChatGPT 3.5 Chatbot with sentiment analysis, and few-shot meta-learning.
- **Keywords:** *LLM Agents, Sentiment Analysis, Few-shot Learning, Meta-learning, MLOps*

GDBWat: Graph DB Watermarking UTD
Database Research Project [Report] Sep 2024 – Dec 2024
Instructor: Dr. Jalal Omer

- Implemented graph database watermarking method for data integrity and ownership verification.
- **Keywords:** *Graph Neural Networks, Data Security, Knowledge Graphs, Graph Database*

Skills

Areas: Computer Vision, Diffusion Models, Robotics, Multimodal AI, NLP, Reinforcement Learning, Transformers, AI Agents, RAG, LLMs, VLMs

AI Frameworks: PyTorch, TensorFlow, Hugging Face, OpenCV, LlamaIndex, LangChain, Pandas, PySpark, SpaCy, ChromaDB, Dify, Copilot Studio

DevOps & Languages: Python (5 years), Java (2 years), C/C++ (2 years), SQL, R, Bash, JavaScript, Git, MySQL, MongoDB, Neo4j, Docker, Flask, OpenShift, Jenkins, Redis

Coursework: Computer Vision[†], Robot Manipulation and Navigation[†], Natural Language Processing[†], Machine Learning[†§], Artificial Intelligence[†], Statistical Methods for Data Science[†], Deep Learning Specialization[‡§], Data Structures and Algorithms[†§]

([†]Graduate level course, [‡]Undergraduate level course, [§]MOOC)

Activities

Indian Institute of Technology, Dharwad Dharwad, India

Institute Music President Aug 2018 – Jul 2020

Music Club Contingent Leader - Inter-IIT Cultural Meet 2019 Dec 2019

Event Head Coordinator - Crescendo 2019 (Live Music Event) Aug 2019 – Sep 2019